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AN OVERVIEW OF THE DESIGN OF DISASTER RELIEF SHELTERS

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Abstract

Disaster relief (DR) shelters play a vital role in large-scale disasters and are an important part of disaster response and recovery. DR shelters are used to provide private and secure places for people to live who have left or lost their usual accommodations as a result of some form of disaster. DR shelters not only provide immediate and short-term shelter for the victims of a disaster, but they also help them to recover from the trauma of a disaster as well as provide a base to start the process of rehabilitation. A review of the literature, case studies, guidance, and reports relating to the design of DR shelters indicates that their provision and performance are not currently as effective as they could be. A lack of adequate consideration with regard to climatic conditions, locally available materials and skills, cultural and social issues, delays, cost constraints, and poor location selection for DR shelters have each been identified as sources of poor performance contributing to an unacceptable standard of living. Moreover, there seems to be a lack of sufficient consideration with regard to the design of DR shelters for future storage and re-use. The principal aim of this research is to examine the extent to which environmental, economic, technical, and sociocultural criteria affect the provision and performance of DR shelters, and how such factors might be taken into account in the decision-making and design processes of such shelters.

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1. Introduction

The research presented in this paper is concerned with disaster relief (DR) shelters. Meeting shelter needs pre- and post-disaster remains a major challenge for governments, humanitarian agencies, and, most important of all, survivors. Disaster shelters are considered vital for personal safety, climate protection, security, and resistance to disease and ill health (IFRC/RCS, 2013). Such shelters are commonly used until a displaced group of people can be re-housed in either their rehabilitated original dwellings or new permanent accommodations. Typical examples of DR shelters include plastic sheets, tents, prefabricated units, and public community buildings such as leisure centres, university halls of residence, places of worship, sports venues, and private rentals. A shelter location may be required for periods that extend to several months or even years after a disaster. Therefore, numerous factors should be taken into account when planning and designing shelters, such as their physical location and the wants and needs of likely users. Complementary support to shelters should come from all relevant stakeholders (e.g., local communities, Non-Governmental Organisations (NGOs), local politicians, and volunteers).

Although the provision of shelters is widely accepted as a necessary component of response and recovery following disasters such as earthquakes, hurricanes, tsunamis, and floods, it is not yet clear which type of shelter is most appropriate given various circumstances that can occur in practice. As a result, the provision and performance of shelters in certain cases has been hindered by inappropriate climate, cultural differences, poorly located settings, camp-related social issues, expenses, overcrowding, poor services, and delays (Barakat, 2003, Nigg et al., 2006, Johnson et al., 2006, El-Anwar et al., 2009, Félix et al., 2013b). In addition, the design of shelters may potentially overlook locally available skills and materials (Johnson, 2007b, Hadafi and Fallahi, 2010), and shelters may not provide an acceptable standard of living. Lastly, in some cases, it has also been difficult to recover shelters for future storage and re-use (Arsalan and Cosgun, 2007).

Therefore, this research seeks to develop clear guidelines for the design of DR shelters by identifying the main environmental, economic, technical, and sociocultural factors to improve the provision and performance of such shelters. The design guidance developed in this research is intended for use by anyone who is involved in the design processes of such shelters.

2. Disaster Relief (DR) Shelters

Adequate shelter has a significant impact on human survival in the initial stages of a disaster (The Sphere Project, 2011). A shelter requires more than just a roof for a space to be habitable. People living in a shelter must have enough clothing, blankets, mattresses, stoves, fuel, and access to services such as water and sanitation (Ashmore, 2004). DR shelters are commonly roofed, secure, hygienic, and liveable locations for people to utilize during periods of disaster until they are able to move back to their permanent dwellings. Many DR shelters are designed and planned so that they can be erected, dismantled, and stored for future use (Arslan, 2007). These kinds of shelters are lightweight structures that can be used for a several purposes (AGOTS, 2007). DR shelters include plastic sheets, tents, prefabricated housing, and public community buildings such as leisure centres, university halls of residence, places of worship, sports venues, and private rentals.

A shelter and a house have different purposes. While shelters offer a safe and secure area to live within immediately following a disaster, houses include daily household responsibilities and work routines (Johnson, 2007b, Félix et al., 2013a). According to UN/OCHA/ESB (2006) and Hadafi et al (2010), adequate shelter is defined as an "immediate environment for all aspects of family life, providing protection from the elements, secure tenure, personal safety, and access to clean water and sanitation, proximity to places of employment and educational and health care facilities." On the other hand, housing is defined as, "Lodging, shelter for human habitation. The immediate physical environment is both within and outside buildings, in which families and households live and which serve as shelter." Housing also includes "government project[s] to provide shelter to low-income groups" (UN/OCHA/ESB, 2006, Hadafi and Fallahi, 2010).

3. Categories of Shelter/Housing

Individuals tend to move between different DR shelter setups before they either return to their previous permanent

residencies, upgrade shelters to permanent house, or build new houses. Shelters can be divided into four categories: emergency shelters, temporary shelters, temporary housing, and permanent housing (Quarantelli, 1991, Wu and Lindell, 2004, Johnson et al., 2006, Johnson, 2007a, Johnson, 2007b, Félix et al., 2013a). However, the International Federation of the Red Cross and Red Crescent Societies (2013) have added additional categories to these, such as transitional shelters, progressive shelters, and core shelters/one-room shelters.

3.1 Emergency Shelter

This type of shelter is used for brief periods of time to deliver life-saving support and is the most basic kind of shelter support (IFRC/RCS, 2013) aside from staying in another permanent building (to be used for a temporary period) for a single night to a few days during an emergency (Quarantelli, 1991, Wu and Lindell, 2004, Johnson et al., 2006, Johnson, 2007a, Johnson, 2007b, Félix et al., 2013a). This kind of shelter commonly does not allow for the extensive preparation of food or prolonged medical services.

3.2 Temporary Shelters

This type of shelter is meant for short-term use. A simple tent or a public mass shelter used for a few weeks following a disaster constitute a temporary shelter (Quarantelli, 1991, Wu and Lindell, 2004, Johnson et al., 2006, Johnson, 2007a, Johnson, 2007b, Félix et al., 2013a). According to the IFRC/RCS (2013), the duration of stay in such shelters may be limited, and therefore, prioritising speed and limiting costs should be taken into account when constructing this kind of shelter.

3.3 Temporary Housing

This type of shelter is often distributed for long-term periods such as six months to three years. Temporary housing such as rental houses and prefabricated unit allow people affected by a disaster to return to their normal daily activities (Quarantelli, 1991, Wu and Lindell, 2004, Johnson et al., 2006, Johnson, 2007a, Johnson, 2007b, Félix et al., 2013a). In many cases, temporary houses are installed on temporary land.

3.4 Transitional Shelters

This type of shelter is usually developed by displaced individuals themselves following a disaster, and such resourcefulness and self-management should be supported (IFRC/RCS, 2013). Transitional shelters are commonly relocated from a temporary site to a permanent location, upgraded into part of a permanent house, resold to generate income to aid with recovery, recycled for reconstruction, and reused for other purposes (International Organization for Migration, 2012). Such transitional shelters are expected to serve for many months or years (Yoshimitsu et al., 2013).

3.5 Progressive Shelters

This type of shelter is designed and built to be more permanent and upgradeable in the future through alterable structural components (IFRC/RCS, 2013).

3.6 Core Shelters/One-Room Shelters

This type of shelter is designed and built with the intent of being permanent housing in the future, including a foundation and all or some of the key services, such as plumbing and various utilities (International Organization for Migration, 2012). The goal with this type of shelter is to build at least one or two rooms to meet permanent housing standards and facilitate improvement. However, these shelters are not intended to be a full permanent house (IFRC/RCS, 2013).

3.7 Permanent Housing

Permanent housing may be upgraded from a transitional shelter, a progressive shelter, a core shelter, or even a new house (Quarantelli, 1991, Wu and Lindell, 2004, Johnson et al., 2006, Johnson, 2007a, Johnson, 2007b, Félix et al., 2013a). Such houses should be resistant and resilient to future hazards and disasters.

Of this range of shelter types, it is best for authorities to understand which type of shelter is most appropriate for a group of survivors' needs and conditions. It is also thought that phases of sheltering and houses are unlikely to work in a neat linear fashion (Quarantelli, 1991, Nigg et al., 2006). For instance, in certain disaster cases, it is recommended to use emergency shelters if damages can be repaired quickly (within weeks) before returning back to one's home, or if one cannot return to his or her own home due to it being too damaged. However, in such a case it would be better to build transitional shelters on one's own land if possible. The earlier the reconstruction process begins, the lower the social and economic costs of a disaster.

4. Issues Related to DR shelters

It is not yet clear which type of shelter is most appropriate for different real-life disaster circumstances. As a result, several environmental, economic, technical, and sociocultural issues can affect survivors when their shelters are designed improperly. Environmental issues arise when changes in climatic conditions are not taken into consideration by designers, such as when simple tents are provided to survivors in a winter season (Johnson et al., 2006, Johnson, 2007a, Félix et al., 2013b), or when a lack of local materials and resources is overlooked (Arslan, 2007, Arsalan and Cosgun, 2007), such as a lack of hygienic water and air (The Sphere Project, 2011), which leads to a significant amount of pollution (Johnson, 2007b).

Economic issues include when a temporary house unit costs more than rebuilding a permanent house. Experts claim that such units can be up to three times more expensive (Johnson et al., 2006, Hadafi and Fallahi, 2010). Another economic issue is a shelter's lifespan. Certain types of shelters, such as temporary houses, are usually set up for a temporary period. However, these shelters often require facilities, services, and utilities such as electricity, sanitation, sewerage, roads, etc. Thus, the entire infrastructure of such shelters requires a significant amount of money, which makes them very expensive to build, especially for less developed and developing countries. Moreover, the delivery of units has been a major issue in the past, as many shelters are manufactured in a different regions or countries and must be imported to the locations in need. (Félix et al., 2013a). Site location is a serious issue for shelters in terms of individuals' proximity to services and livelihood (Camp Coordination/Camp Management, 2010).

Technical issues include lack of space and planning for storing units and materials and illegal occupancy of shelters after a disaster period has passed (Johnson, 2007b). In addition, certain kinds of units are more complex in design than others and require highly skilled workers and kits (Hadafi and Fallahi, 2010, International Organization for Migration, 2012). The performance of shelters tends to suffer when they are small, uncomfortable, and difficult to maintain and upgrade and less of isolations in materials (Arsalan and Cosgun, 2007, IRP and ISDR, 2011).

Finally sociocultural issues include cultural differences between aid suppliers and survivors, which can create misunderstandings, when certain solutions are not suitable for users (Félix et al., 2013b), poor social networks, a lack of places of to communicate, inequality between poor and rich survivors (IRP and ISDR, 2011), lack of support to vulnerable people (e.g., the elderly, disabled, children and women), gender issues, religious issues, and educational issues (Camp Coordination/Camp Management, 2010, International Organization for Migration, 2012).

5. A Review of the Current Guidelines for DR Shelters

Guidance, semi-guidance, and documents on DR shelters have been here analysed in order to understand how such shelters have been constructed in the past, what kind of shelters were involved in certain disasters, and what the environmental, economic, technical, and sociocultural context of the disaster was. For example, the Transitional Shelter Guidance (International Organization for Migration, 2012), Collective Centre Guidelines (Camp Coordination/Camp Management, 2010), and others describe various types of shelters; other guidelines are identified briefly in journal papers, such as "Strategic Planning for Post-Disaster Temporary Housing" (Johnson, 2007b), and "Guidelines to Improve Sustainability and Cultural Integration of Temporary Housing Units" (Félix et al., 2013a), among others; others were written in government documents such as the Evacuation and Shelter Guidance (HM Government, 2014), among others; and lastly, other guidelines can be taken from the Humanitarian Charter and the Minimum Standards in Humanitarian Response (The Sphere Project, 2011), among others.

However, these guidelines do not adequately address certain design aspects of DR shelters to make them more sustainable and comprehensive with regard to different contexts and situations, as well as to solve the many negative issues that have been mentioned above. Design factors define the performance of shelters and should be developed through consultation with the people affected by a disaster, government sectors, private sectors, and any other players involved in disaster recovery, such as volunteers and insurance organisations, to prevent against the environmental, economic, technical, and sociocultural issues listed below, which this research aims to do, to fulfil this need.

1.5 Environmental Factors

1.5.1 Climate Variations

The weather varies significantly between potential disaster locations by season. People of different regions with different temperatures may find different types of shelter more appropriate and comfortable based on their home environment. Design details such as high ceilings and verandas can cause shelters in hot weather to be cooler, and

reducing air gaps or including a lobby area can keep shelters warmer in cold weather (IFRC/RCS, 2013). Temporary, transitional, and progressive shelters should be designed so that they can be winterised (IFRC, 2010). It is not always the case that all people are able to build or repair their own houses.

There are three different types of climatic conditions that each require different shelter arrangements (Corsalis and Vitale, 2005). Survivors should have fuel and stoves available, and they should be protected from the ground by mattresses and beds in extremely cold conditions. Furthermore, they should be supplied with proper clothing, stoves, and blankets. Ventilation is of vital importance in hot-dry and hot-humid climates as well as shade from the sun (Ashmore, 2004). In hot-humid climates, a proper drainage system is recommended. Shelters for hot climates should also take into account possible temperature drops, particularly at night, in open areas such as deserts, and in areas at a high altitude.

1.5.2 Recycling, upgrading, and disposal

The material of DR shelters should be easy to recycle, upgrade, reuse, resell, and relocate after a shelter is disassembled (International Organization for Migration, 2012). Arslan (2007) describes a shelter as “recycled” when it can be partly or completely reproduced from disassembled materials. How beneficial a shelter will be in terms of providing different functions and how resilient it will be in different conditions is important to consider to make it more environmentally friendly. Therefore, it is strongly suggested that shelters be made of materials that can be recycled, upgraded, and reused instead of those that are simply disposed of after use (Kats and Alevantis, 2003). Shelters can even have a positive effect on the environment if they are designed with a dual purpose in mind. Alternatively, shelters that are hard to upgrade and reuse tend to produce more pollution, consume more resources, and thus cause negative impacts on the environment. According to Arslan (2007), reusable and recyclable shelters maybe useful for students, a new couple, low-income families, and holiday camps. In addition, they may also satisfy the conditions of a home by constructing certain additions for them.

1.5.3 Hygienic (water & air)

Promoting personal and environmental hygiene is required in order to protect people’s health. Water, sanitation, and hygiene infrastructure and facilities have to be adequate on campus. However, each of these is often complex in nature and expensive. On the other hand, each are necessary for the health of survivors (Camp Coordination/Camp Management, 2010). The Sphere Project (2011), which set standards for water supply, sanitation, and hygiene promotion, included the following: design and implement wash programmes for people to wash themselves and their clothes and bedding; ensure that people make the best use of their water in terms of the disposal of human faeces, controlling mosquitoes that carry diseases, and drainage work. Lastly, ways to improve survivors’ nutrition should be considered, such as their ability to store, prepare, and cook food.

2.5 Economic Factors

2.5.1 Type of shelters

Money plays a vital role in disaster response and recovery. It is often a critical element in ensuring design and shelter costs. There are several types of shelters that can be used in disaster responses, such as plastic sheeting, tents, prefabricated units, and permanent buildings (to be used for a temporary period). It has been argued that upgrading and improving shelters is cheaper than moving from phase to phase, such as from an emergency response to temporary shelter to permanent reconstructions (International Organization for Migration, 2012). It would be a good practice to compare the price of different shelters between hosts and affected populations (IFRC/RCS, 2013).

2.5.2 Lifetime

The design and planning of shelters should understand their intended life spans given the standards and conditions of the locations where they are to be built (Johnson, 2007b). Thus, in a shelter’s design process, it is important to take into account that a shelter may be hard to build rapidly and will require a significant amount of money if it must last a long time. Sometimes it is preferable for money to be spent on the repair and development of permanent houses rather than on temporary houses in such a situation (Yoshimitsu et al., 2013).

2.5.3 Livelihood

Livelihood support for shelter users applies for the most part to long-term displacement scenarios (Camp Coordination/Camp Management, 2010). After providing initial shelters to affected people, support groups can assist

locals to begin earning money by helping them to start small businesses. For instance, people began to sell “Tamaki” and friendship bands made out of fishing nets following The Great Eastern Japan Earthquake (Yoshimitsu et al., 2013). In addition, items such as key chains, slippers, and fabric bags were hand-made by women. The psychological recovery process of an affected population can be facilitated by encouraging activities that support life and elevate the socioeconomic status of affected people.

3.5 Technical Factors

3.5.1 Easy to Erect and Dismantle

In order to make shelters easy to erect and dismantle to serve many functions and purposes, they need to be of light weight and have few pieces. Certain types of shelters, such as plastic sheets and tents, are simply erected for a short time span and then dismantled (AGOTS, 2007). If the design of a shelter is complex, it will require more training and resources to build it, leading to potential delays (IFRC/RCS, 2013). Therefore, it is important to ensure that shelters will be created on time when developing them.

3.5.2 Materials and Insulations

A consideration of shelter materials should include their quality, cost, appropriateness, local knowledge of the materials, local available materials, their impact on local markets, and the environmental impact of the materials (IRP and ISDR, 2011). The materials used in construction should avoid environmental pollution of any kind. The materials should cause no harmful emissions, and ideally should be made of recyclable, sustainable materials; they should also be easy to manufacture and construct, as well as light in weight. For example, prefabricated wood panels should be 100 kg in weight at most and 3mx1m in dimension to make horizontal and vertical transportation easy. The number of total components should be limited (Sever and Altun, 2009). Furthermore, it is widely agreed upon that shelters should be designed with materials for noise insulation, temperature insulation, and weather insulation.

3.5.3 Classification of Hazards and Performance

A shelter’s structure should be designed so as to protect its occupants from hazards such as earthquakes, storms, and diseases (Camp Coordination/Camp Management, 2010). For instance, it is logical to build timber and/or bamboo-framed structures for earthquake disasters, as such frames are light in weight and thus less likely to cause fatalities than falling masonry structures. On the other hand, in strong winds, these light frames can be more vulnerable (IFRC/RCS, 2013). Furthermore, when a shelter will become part of a permanent home should also be taken into account in areas prone to floods and high winds.

3.5.4 Physical and Psychological Effects

According to Carlier et al (1997), individuals whose homes have been damaged entirely commonly have serious stress issues. Losing a house has a considerable psychological impact and can cause physical stress symptoms. Losing a house can not only cause serious trauma, but can also cause long-term negative outcomes such as post-traumatic stress disorder. Moreover, users’ attitudes and behaviours toward different types of shelters have a crucial impact on levels of post-traumatic stress. Therefore, it is important to consider design elements to reduce stress of affected people when designing and arranging DR shelters (Caia et al., 2010). For example, a sloped roof may be more fitting than a flat roof and large windows.

4.5 Sociocultural Factors

4.5.1 Cultural Difference

Shelter orientations, styles, and design details are different between regions, countries, and even ethnic groups within countries (International Organization for Migration, 2012). As a result, they must be adapted to local communities and their cultures. Providers or aid suppliers must respect and understand users’ cultures to provide adequate shelter solutions for them (Félix et al., 2013a). Shelter solutions must reflect the needs and requirements of users’ traditional values, religions, family sizes, genders, and local architectural styles.

4.5.2 Dignity and Security

Dignity and security in a shelter have a significant impact on individuals that varies from region to region, community to community, and culture to culture. Shelters should not be developed as a simple physical structure, but so as to make users feel socially integrated and provide space to live with dignity and security (IFRC/RCS,

2013). The standards for designing shelters must consider the privacy, dignity, and security of users, as well as encourage flexibility in design, such as the ability of users to add partitions in shelters to achieve greater privacy (International Organization for Migration, 2012). In many circumstances, additional features may be needed in a shelter, such as lockable doors and windows, to provide a basic level of security.

4.5.3 Communication

Communication in the early stages of disaster recovery has a significant impact on survivors, and their participation in recovery decisions during these periods can reduce negative impacts and help them to think about requirements such as their future living locations and needs (Hadafi and Fallahi, 2010). There are several possible means of communication for a community to develop strong social networks: television, radio, internet, mobile phone, newspapers, leaflets, posters, information packs, committees, workshops, and training (International Organization for Migration, 2012).

6. Conclusion

In the event of a disaster, shelters need to be provided as quickly as possible for displaced populations, as losing a house means more than just physical deprivation. Losing a house also implies losing one's dignity, identity, and privacy, which in turn increase illness and pollution. It appears that certain guidelines for DR shelters need to better deal with environmental, economic, technical, and sociocultural issues related to such shelters in order to improve the living quality and needs of displaced people. Therefore, before considering providing shelters for survivors, stakeholders should consider and analyse the design factors of DR shelters to ensure that they are fit for their intended purpose.

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